EDNS 491 & 492: Capstone Design I & II

Student Guide

Academic Year 2025–2026

Adam Duran · Assistant Teaching Professor · PE, PMP
Department of Mechanical Engineering
Colorado School of Mines

How to use this guide: This document is your sprint-by-sprint companion for Capstone Design. It is *not* a replacement for the Course Syllabus, the Course Text, or Canvas; it is a navigation aid that tells you where to focus your energy each sprint. For each sprint, start by reading that sprint's section and pay special attention to any box labeled "Common Pitfall" or "Pro Tip" (those address the most frequent issues). Read the relevant section at the *start* of each sprint, not the end.

Three documents work together: (1) this Student Guide (sprint-level navigation), (2) the Course Text (deep reference, 22 chapters across 5 parts), and (3) Canvas (assignments, rubrics, deadlines). The Course Syllabus is the authoritative source for grading and policies.

A note on numbering: section numbers in this Student Guide (e.g., "Section 3") refer to sections *within this guide*, not to chapters in the Course Text. When referencing the Course Text, this guide always uses the format "Course Text Ch. X" to avoid confusion.

Contents

1 The Capstone Sprint: A Typical Week **4**

2 Course Philosophy and the Three-Document Ecosystem **4**
2.1 How SDI and SDII Fit Together . . . 5
2.2 Your Three Documents . . . 5

3 The Most Common Mistake at Each Sprint **5**

4 Understanding Your Project Track **6**
4.1 Paper Track Projects . . . 6
4.1.1 What a Paper Track Project IS and ISN'T . . . 6
4.1.2 Sprint-by-Sprint Paper Track Adaptations . . . 7
4.1.3 Paper Track V&V . . . 8
4.1.4 Paper Track Deliverable Checklist . . . 8
4.2 Competition Track Projects . . . 9
4.2.1 How Competition Track Differs . . . 9
4.2.2 The Dual Timeline . . . 9
4.2.3 Competition Track Sprint Adaptations . . . 9
4.2.4 Travel and Logistics . . . 10
4.2.5 Competition Team Roles . . . 11
4.3 Hybrid Projects . . . 11

5 SDI Sprint 0: Project Expo, Team Formation, Client Kickoff **11**
5.1 Deliverables . . . 12

6 SDI Sprint 1: The Project Plan, Scrum Training, Client Meeting Prep **12**
6.1 Deliverables . . . 13

7 SDI Sprint 2: Concept Generation, Trade Study, Career Day **13**
7.1 Deliverables . . . 14

8 SDI Sprint 3: Concept Development, Client Approval of Approach **14**
8.1 Deliverables . . . 14

9 SDI Sprint 4: Functional Prototypes / Site Plans, Ethics Assignment **14**
9.1 Deliverables . . . 15

10 SDI Sprint 5: PDR Preparation, Mock PDR, PDR Execution **15**
10.1 Deliverables . . . 16

11 SDI Sprint 6: Design Refinement, Next Steps, SDII Handoff **16**
11.1 Deliverables . . . 16
11.2 Preparing for SDII . . . 17

12 SDII Sprint 7: Re-engagement, Goal Setting, Hazard Assessment **17**
12.1 Deliverables . . . 17

13 SDII Sprint 8: CDR Report, Eng Calc Package, CDR Execution **17**
13.1 Deliverables . . . 18

14 SDII Sprints 9–11: Build/Test/Iterate, Broader Impacts Essay **18**
14.1 Deliverables . . . 19

15 SDII Sprints 12–13: FDR Preparation, System V&V, Final Testing **19**
15.1 Deliverables . . . 19

16 SDII Sprint 14: FDR, Showcase, Documentation, Final Checkout **20**
16.1 Final Deliverables . . . 20

17 Design Review Preparation Checklist **20**

18 What Your PA Expects **21**

19 The Feedback Loop Survival Guide **21**

20 CFO and Purchasing Survival Guide **22**

21 Travel Planning Quick Reference **23**

22 Resume and Career Day Preparation **23**

23 Module Checklist: Sprint-to-Chapter Mapping **23**

24 Quick Reference: Capstone Abbreviations **24**

25 Emergency Contacts and Key Resources **25**

A Assignment and Rubric Reference **26**
A.1 EDNS 491 — Senior Design I Assignments . . . 26
A.1.1 Sprint Cycle Assignments (Sprints 0–5) . . . 26
A.1.2 Project Kickoff & Client Meeting . . . 27
A.1.3 Project Plan (Statement of Work) . . . 27
A.1.4 Preliminary Design Report (PD Report) . . . 27
A.1.5 Preliminary Design Review (Presentation) . . . 27
A.1.6 Ethics Assignment . . . 28
A.1.7 Engineering Standards . . . 28
A.1.8 Project Goal Setting & Goal Attainment . . . 28
A.1.9 Individual Performance Evaluations . . . 29
A.1.10 Next Steps Email . . . 29
A.2 EDNS 492 — Senior Design II Assignments . . . 29
A.2.1 Sprint Cycle Assignments (Sprints 7–14) . . . 29
A.2.2 Re-engagement Email & Activity . . . 29
A.2.3 Critical Design Report (CDR) . . . 30
A.2.4 Critical Design Review (Presentation) . . . 30
A.2.5 Final Design Report (FDR) . . . 30
A.2.6 Final Design Review (Presentation) . . . 30
A.2.7 Broader Impacts Essay . . . 30
A.2.8 Design Showcase (Poster) . . . 31
A.2.9 Project Synopsis . . . 31
A.2.10 Digital Design Archive . . . 31
A.2.11 “What We Wish We Would Have Known” Video . . . 31
A.2.12 Manufacturing Readiness Review . . . 31
A.3 Rubric Reference . . . 31
A.3.1 Preliminary Design Report Rubric . . . 31
A.3.2 Preliminary Design Review Presentation Rubric . . . 32
A.3.3 Critical Design Report Rubric . . . 33
A.3.4 Critical Design Review Presentation Rubric . . . 33
A.3.5 Final Design Report Rubric . . . 34
A.3.6 Final Design Review Presentation Rubric . . . 35
A.3.7 Ethics Assignment Rubric . . . 35
A.3.8 Broader Impacts Essay Rubric . . . 35
A.3.9 Project Plan (Statement of Work) Rubric . . . 36
A.3.10 Design Showcase Poster Rubric . . . 37
A.3.11 Individual Performance Evaluation Rubric . . . 37
A.3.12 Next Steps Email Rubric (SDI) . . . 38
A.4 Grade Weights . . . 39
A.4.1 EDNS 491 — Senior Design I . . . 39
A.4.2 EDNS 492 — Senior Design II . . . 39

1 The Capstone Sprint: A Typical Week

This course meets formally on Tuesdays and Thursdays only (lectures and recitations). The Monday, Wednesday, and Friday activities shown below are *self-directed* team work sessions that you are responsible for organizing outside of class time. Capstone requires significant independent effort beyond scheduled meetings. Teams that rely solely on Tuesday/Thursday class time will fall behind.

Attendance at every lecture and recitation is mandatory. Key deliverable instructions, client feedback, design review logistics, and PA assessments occur during these sessions. Material covered in lecture and recitation cannot always be recovered from slides alone. Missing a session means missing information your team needs to stay on track.

Your team's weekly rhythm should follow this general pattern (adapt to your specific schedule):

Day	Activities
Monday	*Self-directed*: Team stand-up (15 min). Update task board. Identify blockers.
Tuesday	**Scheduled**: Lecture / Recitation / Lab session. Design, analysis, or fabrication work.
Wednesday	*Self-directed*: Work session. PA check-in (show progress, surface issues).
Thursday	**Scheduled**: Lecture / Recitation / Lab session. Send client status email (biweekly).
Friday	*Self-directed*: Work session. Documentation. End of Week 2: Sprint Review + Retrospective.

The three non-negotiable weekly touchpoints: (1) the stand-up (team coordination), (2) the PA check-in (accountability), and (3) the client communication (professionalism). Everything else flexes around your project needs. Plan for 6–10 hours per week of team work beyond scheduled class time.

2 Course Philosophy and the Three-Document Ecosystem

Capstone Design is not another course; it is a professional practice experience compressed into two semesters. You will define problems, manage a team, communicate with a real client, design and build (or model) a system, verify it against requirements, and present your results publicly at the Design Showcase. The skills you develop here (engineering judgment, project management, client communication, teamwork) are the skills employers value most.

What level of independence is expected?

Capstone demands self-motivation. Unlike lecture-based courses, your progress depends on work you organize and complete *outside* of scheduled Tuesday/Thursday class sessions. Your PA will not chase you for deliverables. Your client will not remind you of deadlines. You and your teammates are responsible for holding each other accountable, scheduling your own work sessions, and maintaining momentum between class meetings. Treat this course the way you would treat your first engineering job: your supervisor sets the objectives, but you manage your own time to meet them.

Are we just building what the client tells us to?

No. Think of your team as a **contracted design firm**, not an order-taking service. The client provides the need; you provide the engineering expertise and technical direction. When information is incomplete, make assumptions, document them, and validate with the client. If the client knew exactly what to build and how, they would not need engineers. Lead the design, bring recommendations (not open-ended questions) to client meetings, and iterate when you learn something new. See **Chapter 1** of the Course Text for the full discussion of this philosophy.

2.1 How SDI and SDII Fit Together

SDI (EDNS 491) is the **definition semester**: you go from project assignment through requirements, concept selection, and Preliminary Design Review. Sprints 0–6 cover the left side of the V-Model.

SDII (EDNS 492) is the **execution semester**: you re-engage with the client, complete the Critical Design Review, build/model your system, test it, and present at the Final Design Review and Design Showcase. Sprints 7–14 cover the bottom and right side of the V-Model.

2.2 Your Three Documents

- **This Student Guide**: tells you what to focus on each sprint.
- **The Course Text**: provides the engineering methodology, process guidance, and technical reference. Organized by topic in 22 chapters across 5 parts.
- **Canvas**: assignment rubrics, submission portals, professional development modules, scheduling, and the Labriola InnoHub enrollment site.

3 The Most Common Mistake at Each Sprint

Sprint	Most Common Mistake	Prevention
0	Starting to design before understanding the problem	Spend Sprint 0 listening, not solving. Ask questions, take notes.
1	SOW with no "Out of Scope" section	Explicitly list what you will NOT do. Get client agreement.

Sprint	Most Common Mistake	Prevention
2	Generating only one concept	Use brainstorming, SCAMPER, morphological chart. Generate ≥15 ideas.
3	Decision matrix rigged to support a predetermined choice	Add sensitivity analysis. Vary weights by ±20%.
4	PoC that does not retire the key technical risk	Identify the risk first, then design the PoC to test it specifically.
5	Skipping the Mock PDR	Schedule the mock 3 days before. Fresh eyes catch problems you miss.
6	Not ordering long-lead components	Order PCBs and specialty parts NOW. They arrive in time for SDII.
7	Skipping the Hazard Assessment	Required before fabrication. PA must approve. No shortcuts.
8	BOM missing fasteners and consumables	Walk through the CAD assembly. Every visible component needs a BOM line.
9–10	Big-bang integration	Test each subsystem individually before connecting one at a time.
11	Leaving all testing to Sprint 12	Start V&V testing NOW. Every week of margin is one redesign cycle.
12–13	Documentation debt	Write report sections during the sprint the work is done, not at the end.
14	Incomplete VCRM at FDR	Every requirement must be dispositioned: Verified, Failed (with disposition), Waived, or Deferred.

4 Understanding Your Project Track

All three project tracks—Build, Paper, and Competition—are held to equivalent standards of rigor, engineering process, and documentation quality. Your track determines the *form* of your deliverables, not the *level of effort* or the *depth of analysis*. A Paper Track team that delivers a validated simulation is held to the same standard as a Build Track team that delivers a tested prototype. A Competition Track team that places last but executes a strong engineering process will outperform a team that wins the competition but cuts corners on documentation and verification.

The Build Track is well-covered throughout the rest of this guide. This section provides dedicated guidance for Paper Track and Competition Track teams, including sprint-by-sprint adaptations, common pitfalls, and deliverable expectations.

4.1 Paper Track Projects

4.1.1 What a Paper Track Project IS and ISN'T

A Paper Track project produces a complete, verified technical package that a qualified engineer could use to *implement* the solution. The deliverable replaces the physical prototype—it does not replace the engineering rigor.

A Paper Track project IS:

- An analytical model, simulation, or computational study
- A feasibility study with quantitative analysis
- An optimization or parametric design study
- A site plan, civil design, or infrastructure analysis
- A policy analysis grounded in engineering data

A Paper Track project is NOT:

- A literature review with calculations appended
- A reduced-effort alternative to building something
- A report *about* engineering without *doing* engineering
- A collection of vendor quotes and spec sheets

4.1.2 Sprint-by-Sprint Paper Track Adaptations

The table below maps key course sprints to their Paper Track equivalents. Sprints not listed follow the standard Build Track guidance.

Sprint	Build Track Focus	Paper Track Adaptation
0–1	Problem definition, stakeholder engagement	Same as Build Track
2–3	Physical test specs, hardware requirements	Requirements include analytical performance targets: convergence criteria, accuracy thresholds, validation benchmarks. Concept generation focuses on modeling approaches, solver selection, boundary conditions.
4 (PoC)	Hardware proof-of-concept	Must demonstrate computational goal attainment: converging simulation, validated sub-model, mesh independence study. "I set up the model" is not a PoC.
5 (PDR)	Preliminary hardware design	Present model architecture, preliminary results, validation plan. Reviewers will ask about assumptions, boundary conditions, mesh sensitivity.
6	Baseline design, long-lead orders	Baseline analytical approach. Begin full-fidelity model development.
7–8 (CDR)	Detailed design, fabrication-ready package	Complete model/simulation with preliminary results. Present validation against experimental data or published benchmarks. Must include sensitivity analysis.
9–11	Fabrication, integration, subsystem testing	Full model development, parametric studies, optimization runs. This is your "fabrication" phase.

Sprint	Build Track Focus	Paper Track Adaptation
12–13	System-level V&V testing	Model verification and validation. Independent checks. Documentation of all assumptions.
14 (FDR)	Prototype demo, final results	Present complete results, visualizations, recommendations. Deliverable is the technical package, not a prototype demo.

4.1.3 Paper Track V&V

What counts as Verification and Validation for a model?

Verification: Does the model solve the equations correctly?

- Mesh convergence or grid independence studies
- Time-step independence checks
- Conservation checks (mass, energy, momentum)
- Unit consistency and dimensional analysis
- Comparison to known analytical solutions

Validation: Do the results agree with reality?

- Comparison to experimental data (yours or published)
- Comparison to published benchmarks
- Comparison to independent analytical methods
- Quantified error metrics (RMSE, percent error, confidence intervals)

Both are required. A model that runs without errors is *not* verified. A model that matches one data point is *not* validated.

4.1.4 Paper Track Deliverable Checklist

Your "complete technical package" must contain all of the following:

1. **Model/simulation files** — documented and version-controlled (Git, not email attachments)
2. **Input parameter documentation** — every input justified with a traceable source
3. **Assumptions log** — every assumption listed with rationale and sensitivity impact
4. **Verification evidence** — convergence studies, conservation checks, analytical comparisons
5. **Validation evidence** — comparison to experimental or published data with quantified error
6. **Results and visualizations** — publication-quality figures with labeled axes and units
7. **Recommendations and limitations** — what the model can and cannot predict, and what should be done next
8. **User guide** — step-by-step instructions to reproduce results, including required software, licenses, and computational resources

Paper Track teams often fall behind because they underestimate the time needed for model validation. Plan to spend at least 30% of your SDII time on V&V, not 100% on running simulations. A beautiful model that cannot be validated is not a successful project.

4.2 Competition Track Projects

4.2.1 How Competition Track Differs

Competition Track teams operate in a unique environment with dual constraints:

- Your requirements come from **two sources**: the course AND the competition rules.
- You manage **two timelines**: course sprints AND competition deadlines.
- Competition placement is **NOT** part of your course grade—process and deliverable quality are assessed on the same standards as every other team.
- You still produce **all course deliverables**: SOW, SDRD, PDRt, CDRt, FDRt.

4.2.2 The Dual Timeline

You must create a single integrated timeline in Sprint 0 and update it every sprint. When course and competition deadlines conflict, competition deadlines take precedence *operationally*, but course deliverables must still be completed on time.

Competition Milestone	Typical Timing	Course Gate Alignment
Registration / team declaration	Early SDI	Sprint 0–1 (SOW)
Design report submission	Mid SDI – Early SDII	Sprint 5–6 (PDR)
Technical inspection / safety review	Mid SDII	Sprint 8–10 (CDR through build)
Qualifying / practice runs	Late SDII	Sprint 11–12 (V&V testing)
Competition event	End of SDII or later	Sprint 13–14 (FDR)

4.2.3 Competition Track Sprint Adaptations

Sprint 0–1:
Read the competition rules *cover-to-cover*. Create a compliance matrix mapping every rule to a design requirement. Identify which rules are design-driving vs. administrative. Start the SOW with both course and competition scope.

Sprint 2–3:
Requirements capture must include both course requirements AND competition rules. Use MoSCoW prioritization: competition safety rules are **Must Have**, scoring criteria inform **Should Have**. Non-scoring rules are still mandatory.

Sprint 4 (PoC):
Demonstrate compliance with the highest-risk competition rule. If the competition has a safety inspection, build toward passing it. Your PoC should retire the biggest technical risk, which is often a competition constraint.

Sprint 5 (PDR):
Present the competition compliance checklist alongside the standard PDR content. Reviewers will ask how you plan to pass tech inspection. Have answers ready.

Sprint 6:
Confirm competition registration. Start travel planning **NOW** if the competition is in spring—Mines travel approvals require an 8-week lead time. Order all competition-specific components.

Sprint 7–8 (CDR):
Competition compliance checklist should be nearly complete. Travel plan must be finalized. Registration confirmed. Present your integrated timeline showing the path from CDR to competition day.

Sprints 9–11:
Build and test against competition criteria. Prioritize passing tech inspection and safety requirements *before* optimizing for performance. A car that passes inspection but is slow still competes; a fast car that fails inspection does not.

Sprint 12–13:
Final competition preparation. Practice runs under competition conditions. Logistics finalized: vehicle reserved, hotel booked, equipment shipping arranged. Contingency plans documented.

Sprint 14 (FDR):
Present results regardless of competition placement. FDR evaluates your *engineering process and deliverables*, not your ranking. A team that placed 15th but executed excellent engineering judgment, documentation, and V&V can earn a higher grade than a team that placed 1st with poor process.

4.2.4 Travel and Logistics

Reference the Travel section of this Student Guide for detailed Mines travel policies. Key timeline for competition travel:

- **8 weeks out**: Submit travel request through Mines approval system
- **6 weeks out**: Book flights/transportation
- **4 weeks out**: Vehicle rental and hotel confirmed
- **2 weeks out**: Equipment shipping arranged (if applicable)
- **Day of**: Execute the plan

International competitions add passport, visa, and customs requirements. See Course Text Ch. 16 for international travel guidance.

4.2.5 Competition Team Roles

Consider assigning a **Competition Manager** role alongside your standard Scrum Master, CFO, and other team roles. The Competition Manager is responsible for:

- Tracking competition rules, deadlines, and rule changes
- Managing registration and team enrollment
- Attending all competition organizer webinars and info sessions
- Coordinating travel logistics with the CFO
- Maintaining the competition compliance matrix

This role prevents competition logistics from falling through the cracks while the rest of the team focuses on design and build.

Do not assume the competition rules are your requirements document. Competition rules tell you what your system must *do*; your SDRD tells you how *well* it must do it and how you will *verify* it. You still need engineering requirements with quantitative acceptance criteria, not just "pass tech inspection."

4.3 Hybrid Projects

Some projects span multiple tracks—for example, a team that builds a prototype *and* enters a competition, or a team that produces both a simulation and a physical test article. If your project is a hybrid, establish the following in Sprint 0:

1. **Clearly define which deliverables serve both tracks and which are unique.** A single SDRD can capture requirements for both streams, but you may need separate V&V plans.
2. **Establish a priority framework**: When resources (time, budget, personnel) are constrained, which stream takes precedence? Document this decision and get agreement from your PA and client.
3. **Watch for scope creep.** Hybrid projects are inherently larger in scope. Resist the temptation to do everything for both streams. Use MoSCoW prioritization aggressively.
4. **Document the agreement.** Your SOW should explicitly describe the hybrid nature of the project, the priority framework, and what "done" looks like for each stream. Get written agreement from your PA and client.

Hybrid projects are the highest-risk track for scope creep. If you find yourself saying "we'll just add this for the competition" or "we should also simulate this," stop and check whether the addition is in your SOW. If not, it requires a formal scope change with PA and client approval.

5 SDI Sprint 0: Project Expo, Team Formation, Client Kickoff

Weeks 1–2 | Course Text: Chapters 1, 2

▶ Top Priorities This Sprint

1. Attend the Project Expo and complete the Project Selection Survey.
2. Meet your team, PA, and client.
3. Prepare for and conduct the Client Kickoff Meeting.
4. Set up the team repository (Teams group, file structure).
5. Begin the Individual Design Log (a personal notebook—physical or digital—where you record design decisions, sketches, calculations, and meeting notes throughout both semesters).

What should I do before the kickoff meeting?

Research the client's organization and industry. Prepare a slide deck for project discovery (you will present these during the Sprint 1 client kickoff). Write 10 open-ended questions. Assign roles: facilitator, scribe, timekeeper. See Course Text Ch. 2 for the full kickoff workflow.

Do not start designing during Sprint 0. Your job this sprint is to *understand the problem*, not to solve it. Listen more than you talk at the kickoff meeting. The first idea is rarely the best idea.

Send a follow-up email to the client within 24 hours of the kickoff meeting summarizing key decisions, open questions, and next steps. Copy your PA. This establishes your professionalism from Day 1.

Competition Track: What to do in Sprint 0

- Read the *entire* competition rulebook before the kickoff meeting.
- Create a preliminary compliance matrix mapping competition rules to design requirements.
- Identify the competition registration deadline and add it to your project timeline.
- Determine if the competition is Build Track or Paper Track (or Hybrid).

5.1 Deliverables

- What to Expect from SD Quiz (Canvas)
- Project Selection Survey (by 11:30 AM on the designated date)
- Prof. Dev. – IP Module, Licensure Module
- Email to client to introduce team with team slide

6 SDI Sprint 1: The Project Plan, Scrum Training, Client Meeting Prep

Weeks 3–4 | Course Text: Chapters 7, 11, 13

▶ Top Priorities This Sprint

1. Develop the Statement of Work (SOW) with scope, deliverables, schedule, budget, and roles.
2. Establish the Product Backlog and first Sprint Backlog.
3. Set up task tracking (Trello, Planner, or GitHub Projects).
4. Complete Scrum training modules on Canvas.
5. Create the Team Charter.

How detailed should the SOW be?

Detailed enough that all parties agree on what "done" looks like for every deliverable. Include explicit "Out of Scope" items. Your Project Advisor (PA) evaluates the project against the SOW. See Course Text Ch. 7 for the complete SOW structure.

The most common Sprint 1 mistake is skipping the "Out of Scope" section. If you don't document what you will *not* do, the client may assume it's included.

6.1 Deliverables

- Statement of Work (SOW)
- Team Charter
- Prof. Dev. – Intro to Scrum, Creating Team Culture Reading
- Client Kickoff Meeting slides and meeting minutes
- Initial Product Backlog
- EHS Safety Training (Canvas module)

7 SDI Sprint 2: Concept Generation, Trade Study, Career Day

Weeks 5–6 | Course Text: Chapters 4, 20

▶ Top Priorities This Sprint

1. Generate ≥15 concepts using brainstorming, SCAMPER, and morphological charts.
2. Screen concepts with Pugh matrix; advance top 3–5 to decision matrix.
3. Prepare for Career Day (resume, elevator pitch, professional attire).
4. Sprint 1 Review / Retrospective / Sprint 2 Backlog.

Career Day is a prime networking opportunity. Bring 10–15 printed resumes and practice your capstone elevator pitch. See Course Text Ch. 20 for Career Day preparation.

Do not settle on a concept before generating alternatives. Three variations of the same idea are *not* three concepts. Generate genuinely different approaches before converging.

Track-Specific Requirements Guidance

Paper Track: Include quantitative analytical targets (convergence criteria, accuracy thresholds, validation benchmarks). "Develop a model" is not a requirement; "Achieve mesh-independent results within 5% of published experimental data" is.
Competition Track: Map every competition rule to an engineering requirement with quantitative acceptance criteria. Safety rules = MUST HAVE. Scoring criteria = SHOULD HAVE.

7.1 Deliverables

- Concept generation evidence (sketches, morphological chart)
- Prof. Dev. – Adv. Scrum Training
- Sprint 1 Review / Retrospective documentation
- Project Plan emailed to client

8 SDI Sprint 3: Concept Development, Client Approval of Approach

Weeks 7–8 | Course Text: Chapters 3, 4, 17

▶ Top Priorities This Sprint

1. Complete the weighted decision matrix with sensitivity analysis.
2. Present top 2–3 concepts to client; obtain approval of selected approach.
3. Draft the SDRD with initial requirements.
4. Identify applicable engineering standards.
5. Sprint 2 Review / Retrospective / Sprint 3 Backlog.

What if the client doesn't like any of our concepts?

Ask which aspects they do like and which concern them. Use their feedback to generate a hybrid concept or explore a new direction. Client approval is a scope commitment; invest the time to get it right.

8.1 Deliverables

- Decision matrix with sensitivity analysis
- Client approval of proposed solution (documented in meeting minutes)
- Engineering Standards identification
- Ethics Assignment / Peer Evaluation #1

9 SDI Sprint 4: Functional Prototypes / Site Plans, Ethics Assignment

Weeks 9–10 | Course Text: Chapters 5, 12, 22

▶ Top Priorities This Sprint

1. Build proof-of-concept prototype or develop analytical model.
2. Complete the SDRD (all requirements with TAID and MoSCoW).
3. Apply CLO 5 (sustainability/societal impact) sustainability tool (LEED, ENVISION, DFE, or ESJ). See Course Text Ch. 5, Table "Choosing the Right CLO 5 Tool" for guidance on which tool fits your project.
4. Complete the FDVSE value assessment (Feasibility, Desirability, Viability, Sustainability, Ethical Responsibility). See Course Text Ch. 5, Section 5.2 for the framework and guiding questions.
5. Complete the Ethics Assignment.
6. Sprint 3 Review / Retrospective / Sprint 4 Backlog.

Build Track: the PoC must demonstrate at least one physical function. "We drew it in CAD" is not a PoC. Paper Track: the PoC must demonstrate computational goal attainment (e.g., converging simulation, validated sub-model).

If using InnoHub shops, start machine training *now*. Training slots fill up quickly and you cannot use equipment without completing the required training on the Labriola Canvas site.

9.1 Deliverables

- Proof-of-Concept demonstration (photos/video)
- Ethics Assignment (1000–1500 word argumentative essay)
- Complete SDRD draft

10 SDI Sprint 5: PDR Preparation, Mock PDR, PDR Execution

Weeks 11–13 | Course Text: Chapters 6, 8, 18, 19

▶ Top Priorities This Sprint

1. Finalize PDR Report (PDRt) and send to client in advance.
2. Conduct Mock PDR (practice presentation with PA or another team).
3. Deliver PDR presentation (PDRw) to PA, client, and TAs.
4. Populate the VCRM (plan: all entries "Open" with TAID assigned).
5. Sprint 4 Review / Retrospective / Sprint 5 Backlog.

What does a strong PDR look like?

Strong PDRs have: testable requirements, $\geq$3 distinct concepts, a decision matrix with sensitivity analysis, proof-of-concept evidence (not just claims), an FDVSE value proposition statement, a sustainability assessment, and a team that can answer questions about every section. Before your PDR, the entire team must read and understand Course Text Chapters 6 (Verification), 8 (PDR), and 5 (Sustainability and FDVSE). See Course Text Ch. 8 for the full checklist and common failure patterns.

Do not skip the Mock PDR. Teams that rehearse consistently outperform teams that present cold. Schedule the mock at least 3 days before the formal review.

Track-Specific PDR Expectations

Paper Track: Present your model architecture, boundary conditions, preliminary results, and validation plan. Reviewers will probe your assumptions and ask how you know the model is correct.
Competition Track: Include your competition compliance checklist as part of the PDR. Confirm that registration is complete or in progress. Show the dual timeline.

10.1 Deliverables

- PDR Report (PDRt) submitted to PA and client
- PDR Presentation (PDRw) delivered
- VCRM plan
- Individual Performance Eval #1

11 SDI Sprint 6: Design Refinement, Next Steps, SDII Handoff

Weeks 14–15 | Course Text: Chapters 8, 10

▶ Top Priorities This Sprint

1. Incorporate PDR feedback: categorize, assign owners, complete.
2. Baseline the SDRD (freeze; all future changes under change control; see Course Text Ch. 11 for the change control process).
3. Begin procurement of long-lead-time components for SDII.
4. Prepare SDII transition documentation.
5. Send Next Steps Email to client.

Order long-lead-time components *now*. Do not wait until SDII Sprint 8. Custom PCBs take 2–4 weeks. Specialty sensors take 2–4 weeks. Every week you wait in Sprint 6 is a week less margin in SDII.

Competition Track: If your competition is in spring semester, start travel planning NOW. Mines travel approvals require a minimum 8-week lead time. Late travel requests create administrative problems and may not be approved.

11.1 Deliverables

- PDRt Update (incorporating feedback)
- Next Steps Email to client
- Peer Evaluation #2

- End-of-Semester Course Survey
- Sprint 5 Review / Retrospective / Sprint 6 documentation

11.2 Preparing for SDII

The semester break between SDI and SDII is *not* dead time. Before leaving for winter break: ensure all long-lead-time orders are placed, finalize your SDII transition documentation, and confirm your client's availability for the first week of SDII (Sprint 7). Week numbering resets to 1 in SDII. Review the Sprint 7 section of this guide before the first day of classes so you are ready to hit the ground running.

12 SDII Sprint 7: Re-engagement, Goal Setting, Hazard Assessment

Week 1 | Course Text: Chapters 2, 11, 21

▶ Top Priorities This Sprint

1. Send Re-engagement Email to client (5 required elements: reintroduce team, summarize SDI status, propose SDII goals, request meeting, attach Group Reflection).
2. Complete the Hazard Assessment and Safety Plan.
3. Review and update the Product Backlog with client.
4. Set SDII semester goals with PA and client.

The Hazard Assessment and Safety Plan must be approved by your PA *before* fabrication begins. Do not start building without it. See Course Text Ch. 21 for the template and required content.

12.1 Deliverables

- Re-engagement Email and Group Reflection & Action Plan
- EHS Safety Training (SDII module)
- Hazard Assessment & Safety Plan

13 SDII Sprint 8: CDR Report, Eng Calc Package, CDR Execution

Weeks 2–5 | Course Text: Chapters 6 (Verification), 9 (CDR), 14 (Budget), 15 (Purchasing)

▶ Top Priorities This Sprint

1. Finalize CDR Report (CDRt) and Engineering Calculation Package.
2. Send CDRt to client before CDR presentation.
3. Deliver CDR presentation (CDRw) to PA, client, and TAs.
4. Complete BOM with all components ordered or received.
5. Complete InnoHub Manufacturing Review Meeting (required before ordering build materials).

The Manufacturing Review Meeting is a **required checkoff on the purchasing form**. CP will not process material orders until it is complete. Schedule it early; slots fill up around CDR time.

Freeze the design at CDR. After this point, changes require formal change control (see Course Text Ch. 11 for the change control process). Teams that keep redesigning after CDR arrive at FDR with no working prototype.

Track-Specific CDR Expectations

Paper Track: Present complete model/simulation with results. Validation against experimental or published data is required. Sensitivity analysis showing which parameters drive results. All assumptions documented.
Competition Track: Competition compliance checklist should be nearly complete. Travel plan must be finalized. Show readiness for tech inspection or qualifying rounds.

13.1 Deliverables

- CDR Report (CDRt) with Engineering Calculation Package
- CDR Presentation (CDRw)
- Peer Evaluation #1 (SDII)
- Next Steps Email (post-CDR) to client

14 SDII Sprints 9–11: Build/Test/Iterate, Broader Impacts Essay

Weeks 6–11 | Course Text: Chapters 5, 6, 12

▶ Top Priorities This Sprint

1. Fabricate, assemble, and wire your system (Build Track) or develop full-fidelity models (Paper Track).
2. Test subsystems individually *before* integrating (test each subsystem in isolation, then integrate one at a time).
3. Begin system-level integration testing.
4. Write the Broader Impacts Essay (SDII, 1000–1500 words argumentative).
5. Sprint Reviews / Retrospectives every 2 weeks.

The Spring Break Shipping Trap: campus is closed for a full week during Spring Break. If a component fails during testing and you need a replacement, order it *before* break. Plan all critical procurement to arrive at least one week before break starts.

Start subsystem testing as early as Sprint 9. Do *not* wait until Sprint 12 for your first test. Every week of testing margin you have is a week of margin for redesign if something fails.

14.1 Deliverables

- Broader Impacts Essay
- Mid-Semester Course Survey
- Peer Evaluation #2 (SDII) / Individual Performance Eval #1
- Sprint Review / Retrospective documentation (each sprint)

15 SDII Sprints 12–13: FDR Preparation, System V&V, Final Testing

Weeks 12–14 | Course Text: Chapters 6, 10, 18

▶ Top Priorities This Sprint

1. Complete all verification testing and update the VCRM.
2. Disposition any failed requirements (redesign, waiver, risk acceptance).
3. Write the FDR Report (FDRt) including as-designed vs. as-built comparison.
4. Write the O&M manual.
5. Design and submit the Showcase poster for printing (∼1 week before Showcase).

A VCRM with mostly "Open" entries at FDR indicates insufficient testing. Every requirement must be dispositioned: Verified, Failed (with disposition), Waived, or Deferred. See Course Text Ch. 6 for failure disposition options.

Write the O&M manual for a non-engineer. Use photographs, numbered steps, and clear language. The person who operates your system after you graduate may not have your technical background.

Track-Specific Final Preparation

Paper Track: Spend at least 30% of remaining time on verification and validation, not running more simulations. Document all assumptions. Create a user guide for reproducibility.
Competition Track: Finalize logistics (vehicle, hotel, equipment shipping). Conduct practice runs and dry rehearsals. Remember: FDR evaluates your engineering process, not your competition placement.

15.1 Deliverables

- FDR Report (FDRt) submitted to PA and client
- Complete VCRM (all entries dispositioned)

- O&M manual
- Showcase poster (submitted for printing)
- Design-Build-Test-Redesign Slide (a single slide summarizing key iterations: what was designed, built, tested, and redesigned throughout the project)

16 SDII Sprint 14: FDR, Showcase, Documentation, Final Checkout

Weeks 15–16 | Course Text: Chapters 10, 19

▶ Top Priorities This Sprint

1. Deliver FDR presentation (FDRw) to PA, client, and TAs.
2. Present at the Design Showcase (poster + live demonstration).
3. Complete all final deliverables.
4. Final Checkout with PA.

What is Final Checkout?

A closing meeting with your PA to: hand off all project materials, return borrowed equipment, submit the Digital Design Archive, review your Project Goal Attainment, and formally close the project. Schedule it before the last day of class.

16.1 Final Deliverables

- FDR Presentation (FDRw)
- Digital Design Archive (all project files—CAD, code, reports, test data, meeting minutes—organized in the required folder structure; see Course Text Ch. 10)
- "What I Wish I Would Have Known" team video
- Project Synopsis
- Next Steps Email (post-FDR) to client
- Project Goal Attainment (a brief summary demonstrating how the team met—or did not meet—each project goal established in the SOW)
- Peer Evaluation #3 / Individual Performance Eval #2
- Final Course Survey
- Final Checkout with PA

17 Design Review Preparation Checklist

Reference | Course Text: Chapters 8, 9, 10

Before every design review, complete the **Project Health Check** self-assessment (Course Text, Appendix B). Rate your team Green/Yellow/Red on each of the 12 failure patterns. Discuss any Yellow or Red ratings with your PA.

1 week before:

- Report draft complete and under team review.
- Slide deck outline complete with section owners.
- PoC / demo hardware tested and working.

3 days before:

- Report finalized and sent to client.
- Slides complete; first rehearsal conducted.
- Identified and researched likely Q&A questions.

Day of:

- Arrive 15 minutes early; test projector and demo equipment.
- Backup slides on USB drive.
- Assigned Q&A responders by topic.
- Scribe ready to record feedback.

18 What Your PA Expects

Reference | Course Text: Chapters 11, 13, 17

Your PA evaluates you based on both *what* you deliver and *how* you deliver it. Meeting the checklist is necessary but not sufficient.

What separates an A team from a B team?

A teams: communicate proactively (PA hears about problems before they become crises), incorporate feedback aggressively (PDR weaknesses are addressed by CDR), demonstrate engineering judgment (trade-offs are analyzed with data, not guesses), and produce professional-quality documentation (reports read like they were written by engineers, not by students rushing at midnight).
B teams: meet deadlines but do not exceed expectations, address feedback partially, make design decisions without data, and produce adequate but unpolished documentation.

The single most effective way to improve your PA's assessment: **close the feedback loop**. When your PA gives feedback at a sprint review, write it down, address it, and report back at the next meeting: "At the last review you asked about our thermal analysis. Here are the results." PAs notice when their feedback is taken seriously.

19 The Feedback Loop Survival Guide

Reference | Course Text: Chapter 13

Feedback Loop peer evaluations directly affect your individual grade. SDI has 2 Peer Evaluations and 1 Individual Performance Evaluation; SDII has 3 Peer Evaluations and 2 Individual Performance Evaluations. See Canvas for the specific schedule of each.

There are two types of individual assessment: **Peer Evaluations** (completed by your teammates about each other) and **Individual Performance Evaluations** (completed by your PA about you). Both inform your individual grade. Check Canvas for the specific schedule of each.

A pattern of low Feedback Loop scores (below 3.0 out of 5.0) combined with PA observation of non-participation can result in individual grade reduction, PIP initiation, or course failure regardless of team scores.

How to receive good evaluations: attend every meeting on time, complete your tasks by the agreed deadline, communicate proactively when you are behind, help teammates when you finish early, participate actively in discussions (not just silently), and be responsive to messages within 24 hours.

How to give useful evaluations: be specific. "Jordan was great" does not help the PA calibrate. "Jordan completed the firmware ahead of schedule, debugged the I2C communication issue that was blocking the sensor team, and volunteered to write the CDR power analysis section when the original author was overwhelmed" gives the PA evidence to work with.

Self-evaluation: be honest. PAs cross-check your self-rating against peer ratings and their own observations. Over-rating yourself while peers rate you lower signals a lack of self-awareness. Acknowledging a specific weakness ("I was late on two deliverables in Sprint 10 due to competing course deadlines; I should have communicated the conflict earlier") signals maturity.

20 CFO and Purchasing Survival Guide

Reference | Course Text: Chapters 14, 15

The CFO role is one of the most time-consuming in SDII. Here is the essential workflow:

1. **Get PA approval** before *every* purchase.
2. Check the **approved vendor list** on Canvas.
3. Complete the **Manufacturing Review Meeting** before ordering build materials (required checkoff).
4. Submit the correct form to CP: Purchase Form, Shopping List, or Reimbursement Form.
5. Email CP (capstonepurchasing@mines.edu) with: team name/number, date, vendor, form used.
6. Update the **Budget Tracker** monthly and upload to Canvas by deadline.
7. Pick up packages promptly from Labriola 204.
8. Use the Mines **tax exemption certificate** for all purchases (taxes are not reimbursed).

Purchases made without PA approval may result in **personal financial liability**. Always get approval first.

21 Travel Planning Quick Reference

Reference | Course Text: Chapter 16

- PA must approve Go/No-Go with CLT before any spending.
- Prepare a draft budget before meeting with CP.
- Carpooling required; multiple drivers same trip not reimbursed.
- Mileage: Mines rate, from 1500 Illinois, Golden CO 80401. Save Google Maps PDF.
- Meal receipts: itemized, no alcohol (entire receipt rejected).
- Submit all receipts within one month of trip.

22 Resume and Career Day Preparation

Reference | Course Text: Chapter 20

- Update your resume with capstone contributions (quantify: budget managed, components designed, tests conducted).
- Practice your 60-second elevator pitch.
- Prepare 5–7 STAR stories from capstone for behavioral interviews.
- Dress business casual; bring 10–15 printed resumes.
- Connect with PA, client, and TA on LinkedIn after the project.

23 Module Checklist: Sprint-to-Chapter Mapping

Sprint	Weeks	Primary Activities	Course Text Chapters
SDI (EDNS 491)			
0	1–2	Project Expo, team formation, client kickoff	Ch. 1, 2
1	3–4	Project Plan, Scrum training, product backlog	Ch. 7, 11, 13
2	5–6	Concept generation, trade study, Career Day	Ch. 4, 20
3	7–8	Concept development, client approval	Ch. 3, 4, 17
4	9–10	Prototypes / site plans, ethics assignment	Ch. 5, 12, 22

Sprint	Weeks	Primary Activities	Course Text Chapters
5	11–13	PDR preparation, mock PDR, PDR execution	Ch. 6, 8, 18, 19
6	14–15	Design refinement, next steps, SDII handoff	Ch. 8, 10
SDII (EDNS 492)			
7	1	Re-engagement, goal setting, safety plan	Ch. 2, 11, 21
8	2–5	CDR report, eng. calc package, CDR execution	Ch. 6, 9, 14, 15
9–10	6–8	Build/test/iterate, Broader Impacts essay	Ch. 5, 6, 12
11	9–11	System integration, V&V execution	Ch. 6
12–13	12–14	FDR preparation, final testing, documentation	Ch. 6, 10, 18
14	15–16	FDR, Showcase, archive, final checkout	Ch. 10, 19

24 Quick Reference: Capstone Abbreviations

Abbr.	Meaning	Abbr.	Meaning
BOM	Bill of Materials	O&M	Operations & Maintenance
CDR	Critical Design Review	PA	Project Advisor
CF	Course Faculty	PDR	Preliminary Design Review
CLO	Course Learning Outcome	PIP	Performance Improvement Plan
CLT	Capstone Leadership Team	PoC	Proof of Concept
CP	Capstone Purchasing	Prof. Dev.	Professional Development
EHS	Environmental Health & Safety	SDI	Senior Design I (EDNS 491)
ESJ	Engineering for Social Justice		
FDR	Final Design Review	SDII	Senior Design II (EDNS 492)
FMEA	Failure Mode & Effects Analysis	SDRD	System Design Req. Document
ICD	Interface Control Document	SOW	Statement of Work
InnoHub	Labriola Innovation Hub	TAID	Test, Analysis, Inspection, Demo
MoSCoW	Must/Should/Could/Won't	VCRM	Verification Cross-Ref. Matrix
MVP	Minimum Viable Product		

25 Emergency Contacts and Key Resources

Resource	Contact
Emergency	911
Mines Police (non-emergency)	303-273-3333
Capstone Program Director	Dr. Kristy Csavina, kcsavina@mines.edu
Capstone Purchasing	capstonepurchasing@mines.edu, Labriola 204
InnoHub Operations	Julia Roos, jroos@mines.edu
InnoHub General	innohub@mines.edu
EHS	303-273-3316, https://www.mines.edu/ehs/
Writing Center	https://writing.mines.edu/, appointments at mines.mywconline.com
Labriola Info	https://labriola.mines.edu/

A Assignment and Rubric Reference

This appendix consolidates the key assignments and rubrics for both EDNS 491 (Senior Design I) and EDNS 492 (Senior Design II). Use this as a quick reference for understanding expectations, point allocations, and grading criteria. **Canvas remains the authoritative source** for due dates, submission links, and any mid-semester updates to rubrics.

Rubric criteria summaries in this appendix are condensed for reference. Always review the full rubric on Canvas before submitting—some criteria contain additional detail not captured here.

A.1 EDNS 491 — Senior Design I Assignments

A.1.1 Sprint Cycle Assignments (Sprints 0–5)

Sprint Cycles form the backbone of your Scrum-based project management. Each two-to-three-week sprint follows the same structure:

1. **Sprint Planning** — Define sprint goals, populate the Sprint Backlog, assign tasks.
2. **Execution** — Daily stand-ups, design work, client engagement.
3. **Sprint Review** — Present progress to client and PA (slide deck required).
4. **Sprint Retrospective** — Team reflection on process improvements (*kaizen*).

Submission (each sprint): Upload (1) the Sprint Review slide deck and (2) the team retrospective within 24 hours of your Sprint Review meeting.

Sprint Cycles 0–5 are scored at 0 points individually but contribute to your *Team Assessment – Project Progress* grade (10% of course). Your PA evaluates sprint engagement holistically.

The sprint themes in SDI are:

- **Sprint 0** — Engaging with the Problem (client kickoff preparation and meeting)
- **Sprint 1** — Planning Through Ambiguity (90% complete Project Plan)
- **Sprint 2** — Trade Study & Concept Development (concept selection table)
- **Sprint 3** — Defining Your Concept (primary concept with client alignment)
- **Sprint 4** — From Concept to Preliminary Design (prototyping, feasibility evidence)
- **Sprint 5** — Preliminary Design Reviews (PDR presentation and report)

A.1.2 Project Kickoff & Client Meeting

Points	10 (team)
Deliverables	Team Information Slide, Stakeholder Document, Sprint 0 slide deck, follow-up email to client
Key expectations	Prepare a 5-slide deck demonstrating early engagement; send follow-up email within 24 hrs; document next steps and first Sprint Review date

Arrive at the client meeting with thoughtful, open-ended questions about the *problem*, not premature solutions. Print color copies of your Team Information Slide for the client.

A.1.3 Project Plan (Statement of Work)

Points	100 (team) — initial draft is ungraded feedback; updated submission is graded
Audience	Client (primary), PA, Technical Advisors
Key sections	Project summary, scope of work, customer needs, design requirements (tabulated), stakeholders, background research, engineering standards, budget, material handoff, design process, project schedule (Gantt chart), team members, management tools, communication strategies

Do **not** send the initial draft submission to the client. The initial submission is for PA feedback only. The updated submission is shared with the client via email.

A.1.4 Preliminary Design Report (PD Report)

Points	100 (team) — initial submission for PA feedback; updated submission is graded
Audience	Client (stand-alone document)
Key sections	Executive summary, scope & background, design requirements, concept exploration (3+ concepts with diagrams), concept critique (sustainability/equity/risk tool), concept selection, functional/spatial basis, life cycle analysis, detailed drawings, budget, schedule

Submit the initial PD Report to your client *before* the Preliminary Design Review so they can prepare questions. The updated version incorporates feedback from the review and is attached to the Next Steps Email.

A.1.5 Preliminary Design Review (Presentation)

Points	100 (team + individual components)
Format	15–20 min presentation + 15–20 min Q&A
Audience	Client, mentors, PA, faculty, students
Requirements	Use Capstone presentation template; every member speaks and is listed on their slides; business casual or formal attire required

A.1.6 Ethics Assignment

Points	100 (individual)
Format	750–1500 word essay, PDF, Times New Roman or Arial 12pt
Task	Identify a specific ethical dilemma in your Capstone project; analyze using engineering ethical frameworks and relevant Code of Ethics; determine a professional course of action
Key distinction	Must present a genuine *dilemma* (competing values), not a generic safety statement

Required sections: (1) The dilemma, (2) Stakeholder analysis, (3) Alternative courses of action, (4) Application of Canons, (5) Rationale & decision, (6) Measurement, (7) References (IEEE), (8) Optional Appendix A if AI was used.

AI may be used as a "simulator and debater" (stakeholder simulation, blind-spot checks, proofreading) but **not** to write the draft or generate citations. If AI is used, Appendix A (AI Transparency Statement) is mandatory.

A.1.7 Engineering Standards

Points	100 (individual)
Task	Select a relevant standard/regulation module, summarize it, explain how it applies to your project, share findings with the team, and schedule a team discussion with your PA
Deliverables	Summary document uploaded to Canvas and to the team's shared folder

A.1.8 Project Goal Setting & Goal Attainment

Goal Setting (10 pts, team): Early-semester document stating end-of-semester goals agreed upon by client and PA, with a clear definition of done.

Goal Attainment (100 pts, team): End-of-semester document restating goals and providing evidence of achievement (images, videos, links to drawings, etc.). PA evaluates based on reasonable goal-setting and systematic progress through sprint reviews.

If your team does not have a functional prototype, preliminary site drawings, schematics, or equivalent by the end of SDI, the team will not receive an A regardless of other assignment scores.

A.1.9 Individual Performance Evaluations

Eval #1	100 pts, mid-semester
Eval #2	110 pts, end-of-semester (includes semester reflection questions)
Assessed by	Project Advisor, informed by peer evaluations, sprint review observations, and contribution to deliverables
Submission	Upload a short document describing your technical and professional contributions with evidence (links to documents, deliverables)

If your evaluation falls below 70%, your PA will issue a Performance Improvement Plan (PIP). Address it promptly—this is a professional development tool, not a punishment.

A.1.10 Next Steps Email

Points	10 (team)
Audience	Client (CC your PA and Technical Advisors)
Key elements	Thank client, summarize PDR feedback and how it was addressed (action items table), engineering recommendation for next semester, proposed budget, SDII schedule, attach updated PD Report
Submission	Paste email text or state the date sent

A.2 EDNS 492 — Senior Design II Assignments

A.2.1 Sprint Cycle Assignments (Sprints 7–14)

SDII continues the Scrum cadence with Sprints 7–14. The structure mirrors SDI (planning, execution, review, retrospective). Key sprint milestones in SDII:

- **Sprint 7** — Re-engagement with client and team; action plan for semester
- **Sprint 8** — Critical Design Review preparation and delivery
- **Sprint 9** — Post-CDR iteration; first peer evaluation
- **Sprints 10–11** — Build, test, redesign cycle; mid-semester evaluation
- **Sprint 12** — Showcase preparation; "What We Wish We Would Have Known" video
- **Sprint 13** — Final Design Review and Report
- **Sprint 14** — Design Showcase and final deliverables

A.2.2 Re-engagement Email & Activity

Email	Professional re-engagement email to client at start of SDII
Activity	Group reflection on SDI and action plan for SDII improvement

A.2.3 Critical Design Report (CDR)

Points	100 (team)
Key sections	Design narrative, engineering analysis & calculation package (with TA sign-off), risk assessment (FMEA or risk matrix), test plans & procedures (V&V), technical drawings, technical writing
Critical requirement	All calculations must have sign-off from a Technical Advisor or faculty member

The CDR is your "license to build." If your engineering analysis has gaps or lacks TA sign-off, you risk delays in moving to fabrication and testing.

A.2.4 Critical Design Review (Presentation)

Points	100 (team + individual)
Criteria	Context (20%), Content (25%), Process (25%), Presentation (20%), Graphics (10%)
Format	Same structure as PDR; use Capstone presentation template; business attire required

A.2.5 Final Design Report (FDR)

Points	100 (team)
Key sections	Project overview, design approach, engineering analysis, final solution (referencing how design criteria were met or not), project management (budget, WBS, schedule), lessons learned, technical writing
Purpose	Stand-alone document enabling future engineers to continue the work

A.2.6 Final Design Review (Presentation)

Points	100 (team + individual)
Criteria	Context (20%), Content (25%), Process (25%), Presentation (20%), Graphics (10%)
Note	Must be completed *before* the Design Showcase

A.2.7 Broader Impacts Essay

Points	100 (individual)
Format	Argumentative essay, 1000–1500 words, use provided template
Task	Argue how an engineered system related to your project has positively or negatively impacted society, the environment, and/or the economy. Must take a clear thesis and defend it against opposing arguments.
Key distinction	This is an *argument*, not a report. Must include opposing viewpoints and rebuttals.

Required sections: (1) Introduction & thesis, (2) Background, (3) Proponent arguments, (4) Opponent arguments, (5) Rebuttal, (6) Conclusion, (7) References (IEEE), (8) Optional Appendix A if AI was used.

The top 10 essays are sent to a panel of alumni and faculty judges. Winners are recognized at the Awards Breakfast and published in the Design Showcase program.

A.2.8 Design Showcase (Poster)

Poster Draft	25 pts (team) — submit well before printing for PA feedback
Showcase	100 pts (team) — scored by external judges (alumni, industry, faculty)
Poster specs	48 W × 36 H in PowerPoint; min 24pt font in text boxes; 28–32pt recommended
Judging criteria	Poster & display, project overview, design approach, design solution, next steps (each on 1–5 scale)
Attendance	Mandatory; business casual or professional recommended

A.2.9 Project Synopsis

Points	10 (team) — complete/incomplete grading
Purpose	Short summary for the Design Showcase program booklet
Requirement	Professional, concise, complete, fully proofread

A.2.10 Digital Design Archive

End-of-project submission of all project files, documentation, CAD files, code, and deliverables in an organized digital archive for the client and future teams.

A.2.11 "What We Wish We Would Have Known" Video

Team-produced video sharing advice and lessons learned for future Capstone teams. Shown to incoming SDI students.

A.2.12 Manufacturing Readiness Review

Teams working in the Labriola truck bays or Thorson Design Center must complete safety training, hazard assessments, and access contracts before beginning fabrication. EHS sign-off is required for any safety-related work, chemical usage, or painting.

A.3 Rubric Reference

The following tables summarize the grading criteria for the most important assignments. CLO (Course Learning Outcome) rows that do not count toward scores are omitted. Point values shown are the maximum for each criterion.

A.3.1 Preliminary Design Report Rubric

Criterion	Pts	Exceeds	Meets / Adequate	Marginal
Executive Summary	3	Succinct, highlights all key points	May omit components or lack conciseness	Limited info; must read full report

Criterion	Pts	Exceeds	Meets / Adequate	Marginal
Project Scope & Background	5	Thorough problem statement, existing solutions, standards	Generally clear but missing some depth	Insufficient background
Design Requirements	5	Tabulated with verification methods, types, numbering	Table present but incomplete	Minimal discussion
Concept Exploration	40	3+ unique, feasible concepts with diagrams, functional/spatial basis, pros/cons	Fewer concepts or missing functional basis	One idea with little exploration
Concept Critique	15	Actionable design updates from appropriate sustainability/equity/risk tool	Tool applied but results not well summarized	Poor attempt or no tool applied
Concept Selection	10	Analysis-driven recommendation linked to constraints	Some reasoning but gaps in justification	Unsupported opinion
Detailed Drawings	4	Complete, professional set in appendices	Present but incomplete or unprofessional	Not complete nor professional
Budget	3	Discusses allocation, BoM, initial expenses	Present but lacking detail	Missing
Project Schedule	3	Updated Gantt chart referenced in text, milestones discussed	Present but not fully discussed	Only included, not discussed
Technical Writing	10	Clear, concise, one voice, proofread	Generally well written with some errors	Many errors; difficult to read
Formatting & Citations	2	IEEE citations, ToC, consistent formatting	Some formatting or citation issues	No citations; poorly formatted

A.3.2 Preliminary Design Review Presentation Rubric

Criterion	Pts	Exceeds Mastery	Below Mastery
Design Considerations	16	Thorough problem definition; precise goals, constraints, requirements; strong justification of safety, standards, broader impacts	Vague problem/goals; unclear requirements; little attention to safety or standards
Proposed Solutions	20	Advanced engineering concepts with strong analysis and clear documentation; deep understanding of tradeoffs	Concepts applied incorrectly; minimal analysis; little awareness of tradeoffs

Criterion	Pts	Exceeds Mastery	Below Mastery
Engineering Process	20	Sophisticated concepts, in-depth analysis, comprehensive documentation	Underdeveloped concepts; weak analysis
Team Presentation	30	Exceptionally clear, well-organized, engaging; polished delivery	Lacks clarity or organization; unprepared delivery
Communication & Graphics	10	Original, compelling graphics; proper citations; minimal text	Ineffective graphics; excessive text; no citations
Professional Attire	4	All members in business casual/formal	Lack of professional attire detracts from credibility

A.3.3 Critical Design Report Rubric

Criterion	Pts	Exceeds	Meets / Adequate	Marginal
Design Narrative	10	Clear method/logic; narrative guides reader; work concisely summarized	Present but lacking in some areas	Limited attempt; significant improvement needed
Engineering Analysis & Calcs	20	Exceptionally detailed, error-free, TA-approved calc package	Some gaps or questions unanswered	Incomplete or lacking TA sign-off
Risk Assessment	15	Detailed critique with risk tool (FMEA/matrix); clear mitigation plans linked to top risks	Tool present but surface-level analysis	Feels like afterthought; no clear narrative
Test Plans (V&V)	10	Complete, descriptive; non-team-member could execute tests	Adequate detail but needs improvement	Work needs significant improvement
Technical Drawings	15	Professional-quality, properly annotated, complete set	Present but formatting issues or incomplete	Limited attempt at professional quality
Technical Knowledge	5	Appropriate for completion level; clear justification of calculations and design decisions	Adequate but needs improvement	Significant improvement needed
Technical Writing	15	Clear, concise, one voice, proofread	Several errors affecting clarity	Many errors; not senior-level

A.3.4 Critical Design Review Presentation Rubric

Criterion	Pts	Full Marks	Poor
Context	20	Succinctly defines problem; clear goals, constraints; addresses safety and broader impacts	10
Content	25	Appropriate engineering concepts, analysis, documentation; understands design tradeoffs	13
Process	25	Strong critical thinking in selecting design methods; prioritizes stakeholders	13
Presentation	20	Extremely well-organized and well-coordinated; creates excitement	10
Graphics	10	Original, compelling, effective; cites all external images	5

A.3.5 Final Design Report Rubric

Criterion	Pts	Exceeds	Meets / Adequate	Marginal
Project Overview	10	Clear history and chronology; usable by future engineers	Generally clear but missing content	Limited documentation
Design Approach	10	Problem-specific approach with correctly applied tools; meets constraints	Basic approach but lacking clarity	Vague or incomplete methodologies
Engineering Analysis	15	Exceptionally detailed, error-free, no gaps; incorporates prior iteration feedback	Some gaps; questions unanswered	Not up to senior-level standard
Final Solution	30	Fully described; testing demonstrates criteria met; supporting documentation complete	Adequate but incomplete documentation	Solution does not meet criteria
Project Management	5	Accurate, up-to-date budget, WBS, schedule; clear path forward	Present but lacking detail	Substantial omissions
Lessons Learned	5	Thoughtful discussion of deviations and key takeaways	Adequate but could be deeper	Conflicts with PA observations
Technical Writing	15	Clear, concise, one voice	Several errors; some parts difficult to read	Many errors; not senior-level
Spelling & Grammar	5	Error-free	A few errors	Too many for senior writing

Criterion	Pts	Exceeds	Meets / Adequate	Marginal
Formatting & Citations	5	IEEE citations; consistent formatting throughout	Some citation/formatting issues	No citations

A.3.6 Final Design Review Presentation Rubric

Identical structure to the CDR Presentation Rubric: Context (20%), Content (25%), Process (25%), Presentation (20%), Graphics (10%). See Section A.3.4.

A.3.7 Ethics Assignment Rubric

Criterion	Pts	Full Marks	Inadequate
Intro to Ethical Consideration	20	Thoughtful introduction; clearly relates to project; explains why it is an ethical challenge	Limited relevance to project
Affected Stakeholders	15	Thoughtful consideration of parties, interests, responsibilities	Limited discussion
Alternative Courses of Action	15	Thoughtful alternatives with consequences to stakeholders	Limited discussion
Ethics & Liability Elaboration	15	Effectively uses Code of Ethics; discusses liability impacts	Minimal use of codes in context
Rationale for Chosen Path	20	Clear, concise rationale based on ethical frameworks and values	Limited discussion
Technical Writing	10	Clear, concise, no grammatical errors	Many errors; difficult to read
Citations & Formatting	5	IEEE format; all non-general-knowledge cited	No references to Codes of Ethics

A.3.8 Broader Impacts Essay Rubric

Criterion	Pts	Excellent	Needs Improvement / Poor
Relevant	25	Addresses prompt in a controversial, insightful, or novel way	Does not understand intent of assignment

Persuasive	25	Inspiring, thought-provoking; strong evidence of engineering education	Not overly creative; arguments weak
Argumentative Essay Format	10	Well-organized; thesis, proponent/opponent arguments, strong conclusion	Reads more like informative essay or report
Well Written	20	Compact, efficient, concise; could be considered for publication	Rambling, narrative, or noisy writing
Spelling, Grammar, Punctuation	10	Practically no errors; strong evidence of proofreading	Pervasive errors; significant distraction
References & Citations	10	Thoroughly researched; IEEE format; scholarly sources	Little to no evidence of research

A.3.9 Project Plan (Statement of Work) Rubric

Criterion	**Pts**	**Exceeds**	**Marginal / No Evidence**
Project Plan Summary	2	Professional, complete, concise	Missing
Scope of Work	5	Clear, succinct problem statement; constraints well defined	Could be much improved
Customer Needs	5	Clear, detailed list with importance ratings	Not provided
Design Requirements	10	Tabulated (not bulleted); complete; linked to client needs	Incomplete and vague
Stakeholders	3	Clear articulation with roles and timing	Insufficient detail
Budget	2	Discusses allocation and fundraising plans	Not addressed
Background Research	10	Demonstrates excellent understanding; existing solutions discussed	No evidence of understanding
Eng. Standards	9	Identifies standards/regulations with detail on project applicability	One or no standard identified
Material Handoff	3	Solid list of deliverables to client	Limited

Criterion	Pts	Exceeds	Marginal / No Evidence
Performance Criteria	3	Thoughtful consideration of V&V approach	Limited attempt
Design Process	5	Thoughtful discussion of engineering design steps	Limited attempt
Project Schedule	5	Detailed Gantt chart with discussion	Only reports/reviews; missing milestones
Team Members	3	Table with names, skills, roles, contact info, advisors	Poorly constructed
Management Tools	3	Strong management plan articulated	Not addressed
Communication	3	Frequency, mediums, status report strategy	Missing information
Technical Writing	20	Clear, concise, one voice, proofread	Lacks clarity; no attempt to edit
Formatting & Citations	2	IEEE citations; consistent formatting	No citations; poorly formatted

A.3.10 Design Showcase Poster Rubric

Criterion	Pts	Excellent	Poor
Tables, Graphs, Figures	5	Correctly labeled, titled, explained; visuals convey information	Not labeled; visuals do not convey info
Grammar & Spelling	5	Free of all errors	Significant errors; poor editing
Quality of Poster	5	Legible from 5+ ft; appropriate color/font; phrases not sentences	Not legible; poor color/font; text-heavy
Design Process / Terminology	5	Clear presentation of process and final selection; appropriate terminology defined	No design process presented
Logical Organization	5	Clear language and images in logical sequence; includes analysis, intro, conclusion, future work	Illogical sequence; little relevant info

A.3.11 Individual Performance Evaluation Rubric

Criterion	Pts	Excellent	Poor / Unsatisfactory
Technical Contribution	40	Produces excellent work; goes above and beyond; seeks help when appropriate; submits early for review	Poor quality; needs to be assigned work; leaves tasks for others; work often late
Professional Contribution	40	Positive influence; completes all sprint tasks; helps others; demonstrates leadership; always on time; professional with all stakeholders	Unprofessional at times; does not complete sprint tasks; consistently late; unavailable outside class
Assignment Submission	20	Fully addresses technical and professional contributions with clear supporting evidence	Minimally addresses contributions with no evidence

A.3.12 Next Steps Email Rubric (SDI)

Criterion	Pts	Full Marks
PDR Action Items	10	Clear, complete list with status
Project Budget & Plan	10	Complete budget provided; original plan referenced; updates discussed
Next Steps Discussion	10	Clear path forward; client expectations addressed; concerns noted
Technical Writing	10	Clear, concise, one voice
Spelling & Grammar	5	No errors
Formatting & Citations	5	IEEE format; consistent formatting

A.4 Grade Weights

A.4.1 EDNS 491 — Senior Design I

Assignment Group	Weight
Team Assignments — Design Reports	35%
Individual Performance Evaluation	20%
Individual Writing Assignments	10%
Team Assessment — Project Progress	10%
Team Assignments — Design Reviews	10%
Prof. Dev. — Engineering Standards	5%
Prof. Dev. — New Employee Training (Complete All)	3%
Team — Design Modules	3%
EHS Safety Training & Access to Labriola Spaces	2%
Prof. Dev. — Peer Evals, Surveys, Reflections	2%
Total	**100%**

A.4.2 EDNS 492 — Senior Design II

Assignment Group	Weight
Performance Evaluations — Individual	25%
Project Planning & Completion — Team	25%
Individual Writing Assignment	10%
Design Reviews — Team	10%
Critical Design Report — Team	10%
Capstone Design Showcase — Team	10%
Professional Development — Individual	5%
Final Design Report — Team	5%
Final Checkout	0%
Total	**100%**

In SDI, the biggest grade drivers are Design Reports (35%) and Individual Performance Evaluation (20%). In SDII, Performance Evaluations (25%) and Project Planning & Completion (25%) dominate. Prioritize accordingly.

This guide is a companion to the EDNS 491/492 Course Text. For detailed methodology, technical reference, and worked examples, consult the Course Text chapters referenced in each section above.